

INDEPENDENT STUDY PORTFOLIO ASSESSMENT

Creature Design & Production Pipeline

Percy Holway · 3D Character & Creature Artist · UCF



Final_2



This portfolio demonstrates fulfillment of four core learning outcomes through a comprehensive creature design study. Work spans anatomical studies (human hand, alligator gar, deer nose sculpt), a complete production ready herbivore creature from concept through environmental render, and workshop collaboration on Zigmol for Mol's Morsels.

01 — DEMONSTRATE ANATOMICAL MASTERY

Demonstrate Anatomical Mastery

Requirement: Successfully apply advanced knowledge of human and comparative animal skeletal and muscular anatomy to inform a complex creature design.

Human Hand Sculpt

Foundation work demonstrating understanding of human skeletal and muscular anatomy through life observation. Modeled from personal life reference, capturing the complexity of hand biomechanics.

Software: ZBrush iPad)

Reference: Life observation · Anatomy for Sculptors by Uldis Zarins

Video: <https://youtu.be/ctBTcpLhyzl>

RENDER PHOTOS

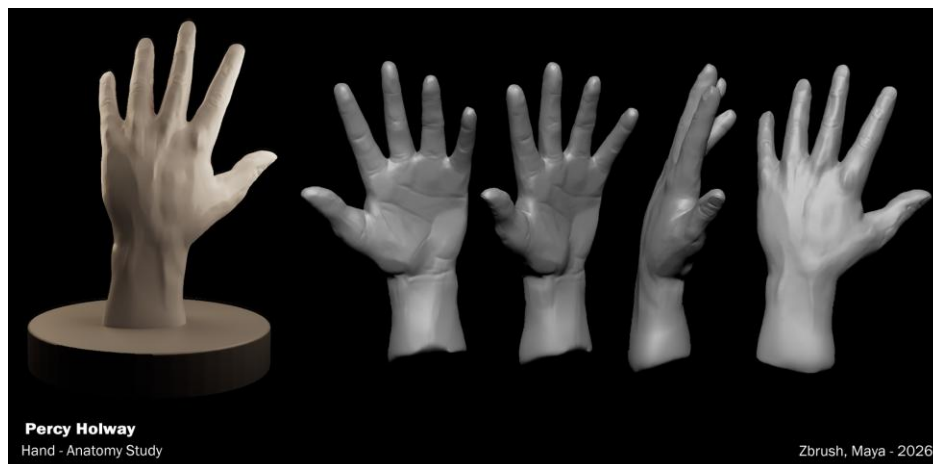


Reference — View 1

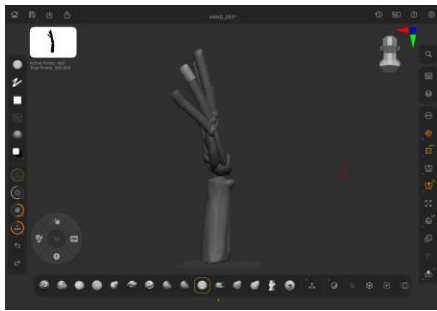


Reference — View 2

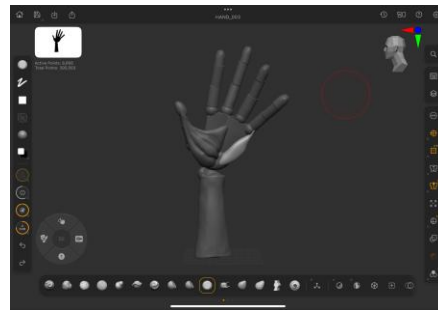
FINAL PORTFOLIO RENDER



SCULPT PROGRESSION



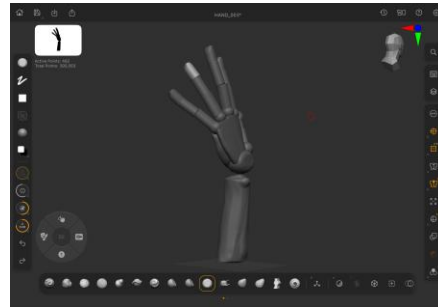
Initial Blockout



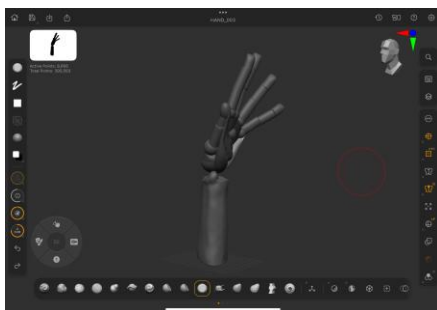
Anatomy Development



Primary Forms



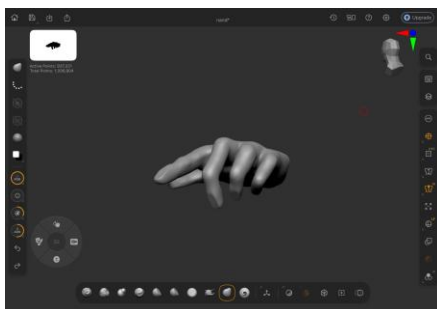
Secondary Detail



Muscle Groups



Tendon & Surface



Initial Sculpt



Attempt 1

"I was modeling this hand by looking at my own. I took a few pictures but was modeling using just observations from life. It's definitely in an ugly stage right now but wow I didn't realize how weird hands were. Getting the proportions right is really tricky."

Deer Nose Sculpt Study — Custom Alpha Development

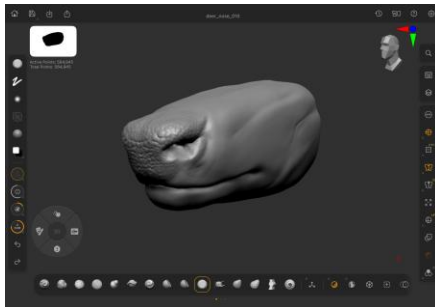
Focused anatomy study of deer nose/snout anatomy — approximately 4 hours. Directly produced the custom alpha brushes to use for the details.

Time: ~4 hours

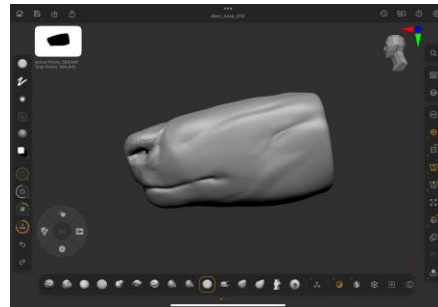
Outcome: Custom alphas derived from sculpt

Software: Zbrush for iPad

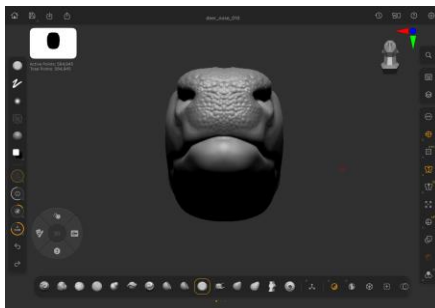
DEER NOSE SCULPT — 4-HOUR STUDY



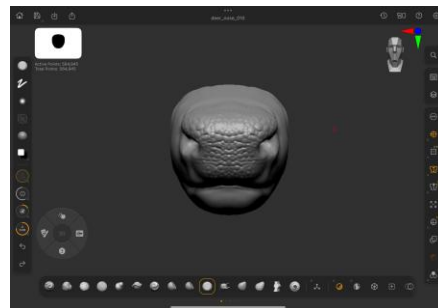
Front View



Side View



Nostril Detail



Surface Texture

"It took me around 4 hours to do the deer nose sculpt as I figured out how to make custom alphas on my iPad. With the restrictions of the iPad, I think I'll be making alphas on desktop as well."

Custom Alpha Creation

Custom alpha brushes developed from deer nose study. Applied throughout the herbivore creature's high-resolution sculpt for skin texture, pore distribution, horn striations, and surface detailing. Learned techniques on alpha placement for realism.

CUSTOM ALPHAS — IPAD & DESKTOP



Custom Alpha 01 — Skin Texture



Custom Alpha 02 — Pore Detail



Custom Alpha 03 — Surface Variation



Custom Alpha 04 — Horn Striation

■ Alligator Gar — Anatomy Study

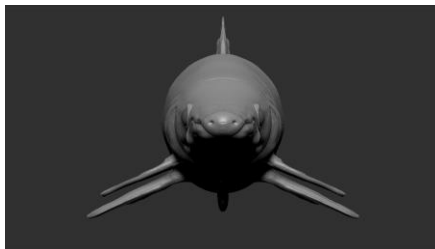
Comparative animal anatomy study of a swamp predator species. Originated as a 4-hour speed sculpt, then pushed to a complete production asset. The alligator gar represents an ancient lineage of fish, providing critical research for predatory adaptations in semi-aquatic environments.

Software: ZBrush · Maya · Substance 3D Painter

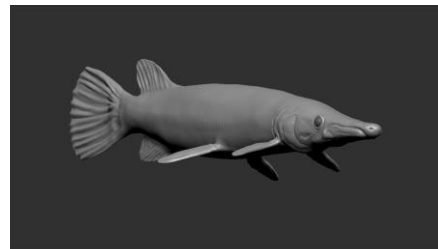
Focus: Elongated skull morphology · scale arrangement · fin anatomy

Application: Scale patterns, fin anatomy, and texturing techniques informed the herbivore's environmental adaptations

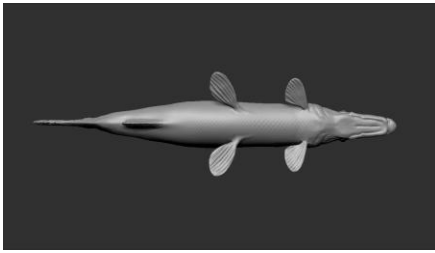
GRAY SCULPT — ANATOMY STUDIES



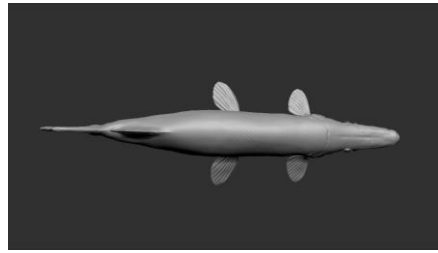
Side Profile — Full Body



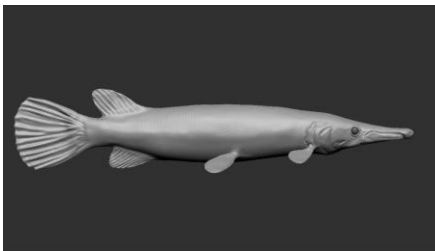
Side Profile — Wireframe Overlay



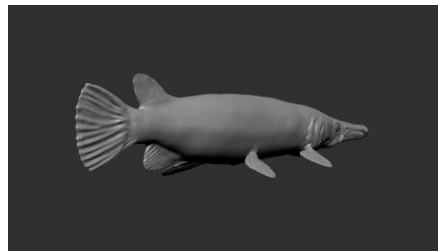
Head Detail



Tail & Fin Anatomy



Ventral View



Front — Snout Detail

FINAL TEXTURED RENDERS — ALBINO ALLIGATOR GAR



Final Posed Render



Head & Face — Texture Detail



Side Profile — PBR Textured



Scale Close-Up



Belly & Front View



Belly View — Final Render

SCULPT REFERENCE



Reference Compilation

"However I did restart the hand and man I just wasn't feeling it so I took a break and sculpted an Alligator Gar for a lot longer than I intended... However I'm gonna keep whittling away at the hand in smaller bursts bc I am determined to get it done."

Herbivore Creature Design — Primary Portfolio Asset

Complex creature combining anatomical features from three swamp-adapted species: tufted deer, capybara, and tapir influences. Demonstrates synthesis of comparative anatomy research into a cohesive, believable organism.

Tufted Deer: Skeletal structure · horn placement · ungulate leg anatomy · alert posture

Capybara: Body proportions · semi-aquatic adaptations · fur texture reference

Tapir / Salamander: Nose morphology · amphibious lifestyle · webbed foot structure

- ▶ Deer-inspired skeletal structure with specialized foot mechanics for marsh terrain
- ▶ Capybara-influenced body proportions optimized for semi-aquatic lifestyle
- ▶ Muscular anatomy adapted for both swimming and terrestrial movement
- ▶ Four-toed webbed hooves preventing sinking in marshy terrain

02 — EXECUTE A PRODUCTION-READY ASSET

Execute a Production-Ready Asset

Requirement: Create a high-resolution creature sculpt and translate it through a complete production pipeline, resulting in a clean, low-poly mesh with optimized UVs.

Herbivore Creature — Concept & Design Development

The herbivore creature design progressed from initial brainstorming through species research, concept sketching, and color reference development before entering the 3D pipeline.

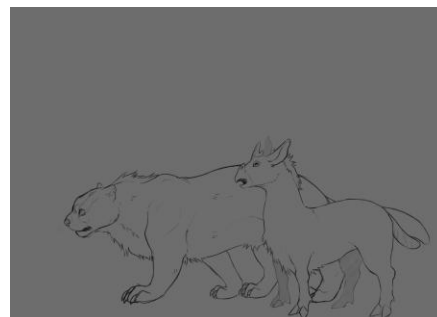
CONCEPT SKETCHES — SPECIES COMBINATION STUDIES



Concept Sketch 01 — Capybara Build with Deer Features



Concept 02 — Alternative Silhouettes



Concept 03 — Form Exploration

SPECIES REFERENCE RESEARCH — DEER & COLOR DEVELOPMENT



Color Research — Tufted Deer & Capybara Reference (with fur color notes)

TEXTURE & COLOR DEVELOPMENT — WIP

Color variation development informed by real-world reference species, exploring White-Tailed Deer and Tufted Deer fur palettes.



Color Variant A — Warm Brown



Color Variant B — Dark Gray



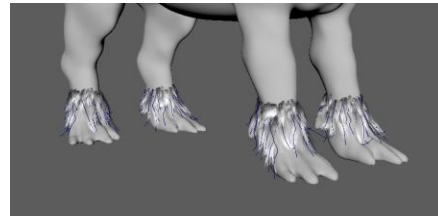
Head Texture WIP



Texture Development



WIP Render



Maya Hair Card — Placement Test

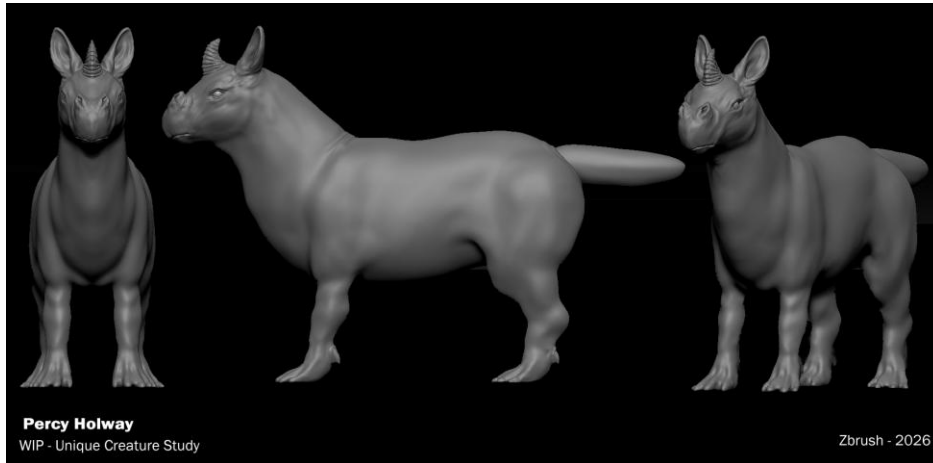


Hair Card WIP

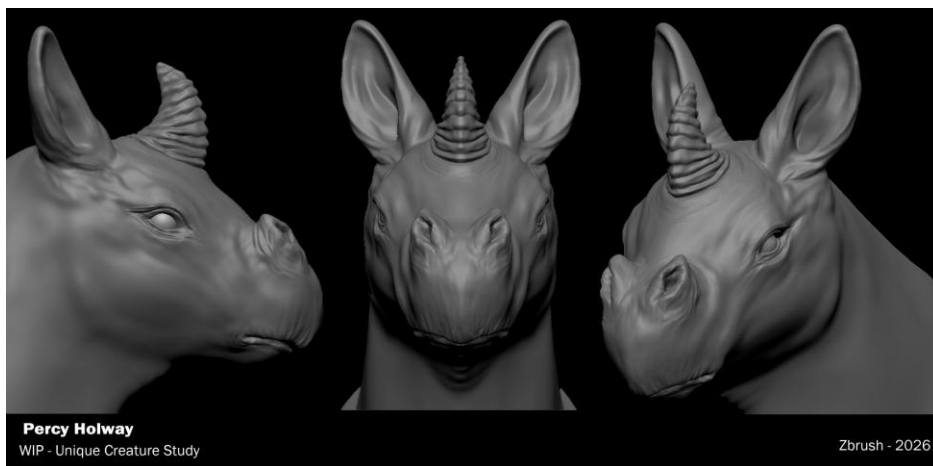


Substance Painter WIP

FINAL PORTFOLIO BLOCOUTS — HERBIVORE CREATURE WIP



Full Body — Latest Render



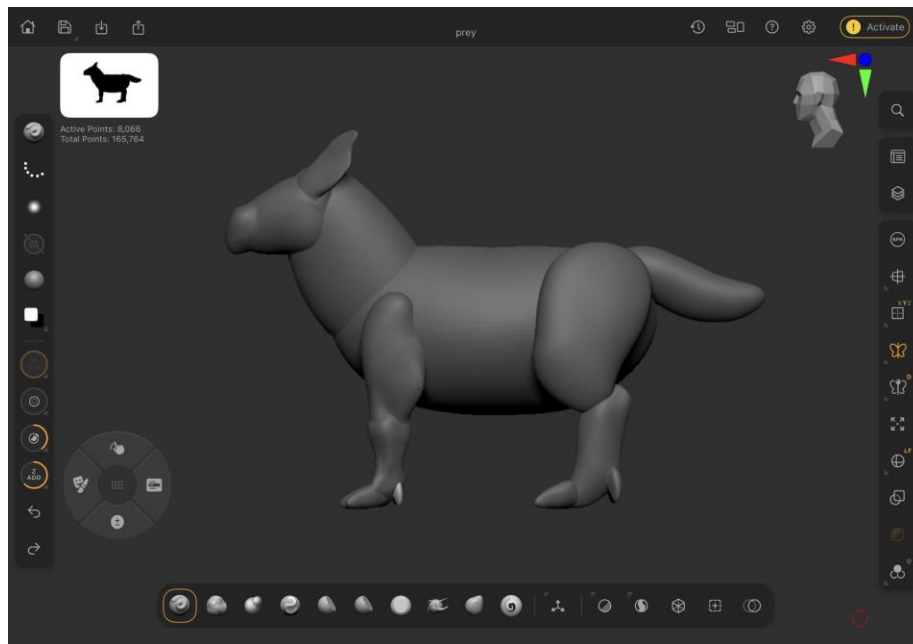
Face & Head Detail

Stage 1 — High-Resolution Sculpting

Software: ZBrush (desktop) · Nomad Sculpt (iPad mobile workflow)

- ▶ Multiple iterations establishing primary forms, secondary anatomy, and tertiary surface detail
- ▶ Horn striations created using custom alphas derived from deer nose study
- ▶ Managed ZBrush license delays via Nomad Sculpt — maintained full productivity

SCULPTING PROGRESSION — PHASE 1: INITIAL BLOCKOUT



Phase 1 — ZSphere Blockout: establishing primary silhouette and mass distribution

PHASE 2: ANATOMY REFINEMENT

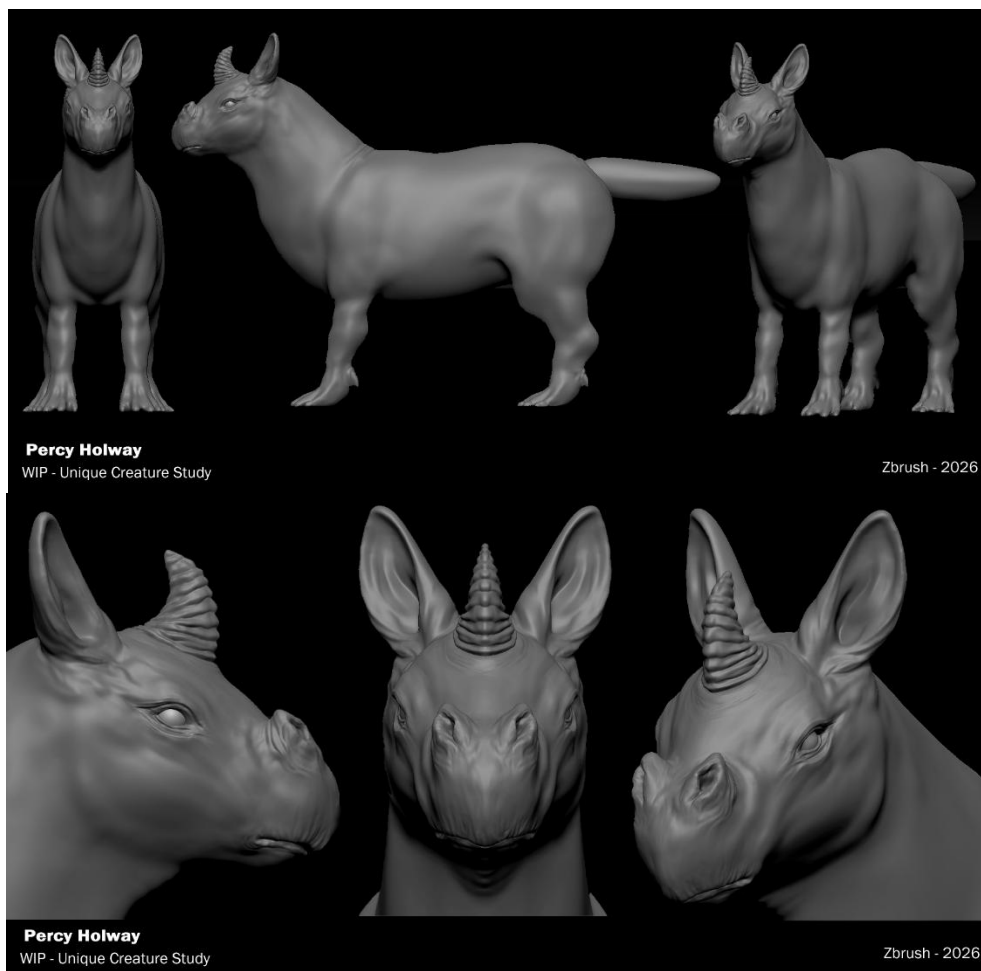


Phase 2 — Secondary forms, muscle groups, horn structure and ear placement defined

PHASE 3: FINAL HIGH-RESOLUTION SCULPT



Phase 3 — Tertiary detail: horn striations, skin pores, eye detail, nose moisture, wrinkles



Percy Holway
WIP - Unique Creature Study

Zbrush - 2026

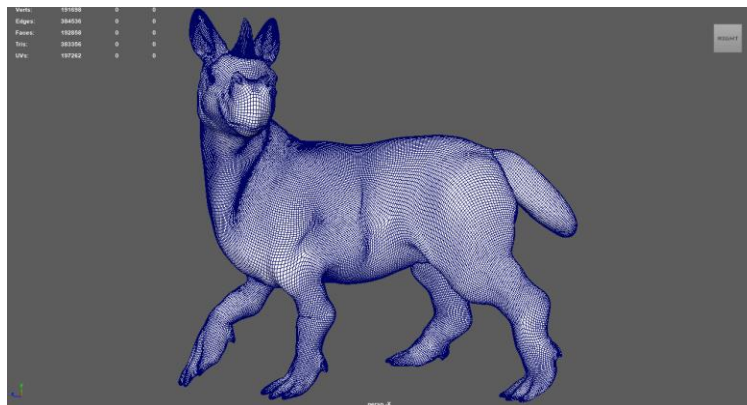
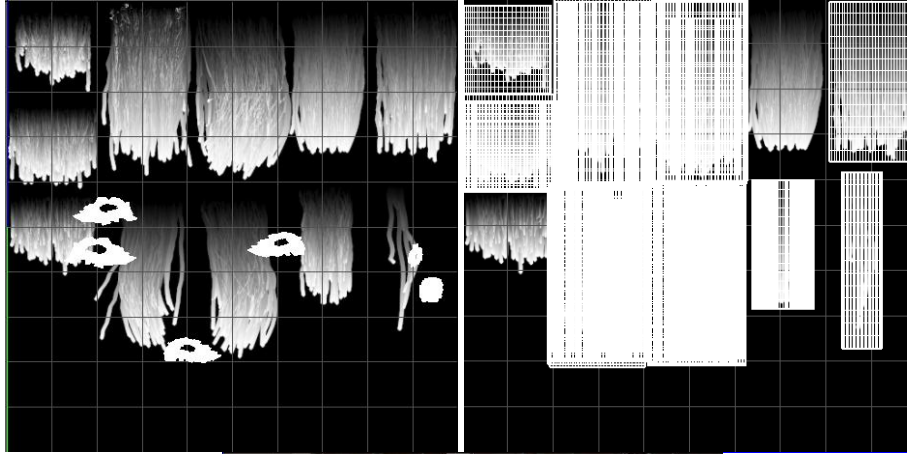
Percy Holway
WIP - Unique Creature Study

Zbrush - 2026

Stage 2 — Retopology & UV Optimization

- ▶ Clean quad-based topology following natural edge flow for deformation
- ▶ Edge loops placed strategically: eyes, mouth, joints, horn base
- ▶ Multiple UV sets: UV Set 0 for textures · UV Set 1 for hair card color projection

UV LAYOUT — OPTIMIZED FOR TEXTURE BAKING



Zremeshed Sculpt — In-Pose

Stage 3 — Texture Baking & Material Development

Software: Substance 3D Painter

Video: <https://youtu.be/a7bY2fN2tGg>

- ▶ Normal Maps: baked from high-res sculpt — skin pores, horn striations, wrinkles
- ▶ Roughness: wet nose · smooth polished horns · textured dry skin
- ▶ SSS: ear backlight transmission, thin skin areas, nose cartilage

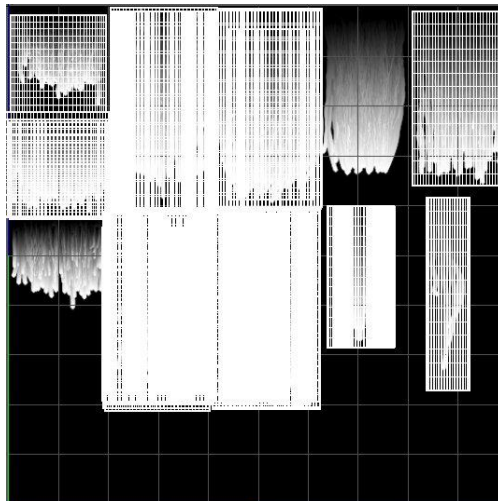
"I did fix the textures from last week making the horn shinier, the ears softer, added more eye texture, and added more transition furs to hide the direction change on the mesh."

Stage 4 — Hair Card System

Pipeline: FiberShop → GS Curve Tools (Maya) → Unreal Engine 5

- ▶ FiberShop: flyaways, thick base strands, shaggy capybara-inspired sections, facial detail
- ▶ 3-Layer Method: dense base → mid-density seam hiding → sparse flyaway silhouette
- ▶ Second UV set for color projection — fur inherits body texture colors

HAIR CARD LIBRARY — FIBERSHOP EXPORT



"Hair cards I believe are done! I have fly aways, thicker base strands, longer shaggier strands, and short strands for the face and legs."

03 — ACHIEVE PHOTOREALISTIC PRESENTATION

Achieve Photorealistic Presentation

Requirement: Generate all PBR texture maps and perform final look development to render the creature in a cinematic presentation.

Albedo: Color variation — natural fur patterns, skin tones, two individual color variants
Normal: Baked from high-res sculpt — skin pores, texture, horn striations, wrinkle depth
Roughness: Wet nose · smooth polished horns · textured dry skin
SSS: Ear backlight, thin skin, nose cartilage translucency
AO: Contact shadow enhancement for depth in recessed areas

FINAL ENVIRONMENTAL RENDER — CINEMATIC PRESENTATION



Photorealistic swamp environment — Unreal Engine 5, FAB marketplace assets, mother and offspring interaction

Rendering: Unreal Engine 5 — FAB marketplace swamp environment
Lighting: Cinematic atmospheric fog, volumetric lighting, god rays, dappled swamp light
Reference: <https://www.artstation.com/artwork/IRP3gY>

04 — MANAGE A COMPLEX TIMELINE

Manage a Complex Timeline

Requirement: Self-direct and manage a 16-week project schedule, utilizing buffer time for unforeseen issues and compiling a final portfolio-ready presentation.

Proactive Schedule Adaptations

- ▶ Workflow Optimization: shifted deadlines to Mondays aligning with closing-shift work schedule
- ▶ Scope Refinement: restructured plan after identifying overscoping issues in workshop group
- ▶ Strategic Pivoting: joined new workshop group (Abi and Sophie) creating productive synergy
- ▶ Tool Management: ZBrush license renewal; Nomad Sculpt as mobile contingency

"I believe I need to move my due dates to Mondays instead of Sundays as I usually work closing on the weekends so having Mondays off and dedicated to workshop stuff I think will be the most helpful."

Parallel Project Management

Independent Study: Herbivore creature — full pipeline from concept through environmental render

Anatomy Studies: Human hand · alligator gar · deer nose sculpt · custom alpha creation

Workshop: 'Zigmol' for Mol's Morsels — sculpt, retopology, UV, rig, weight paint, animation

Professional: Portfolio website, resume, GDC conference preparation, ArtStation updates

Workshop Collaboration — "Zigmol" for Mol's Morsels

Concurrent team-based creature development for a student short film. Full pipeline ownership: ZBrush sculpt → Maya retopology → UV layout → rigging → weight painting → animation testing. Team: collaboration with Abi and Sophie.

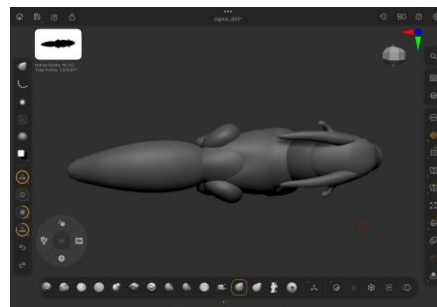
GRAY SCULPT — HIGH-RESOLUTION DETAIL



High-Resolution Sculpt — Top View with Dorsal Details

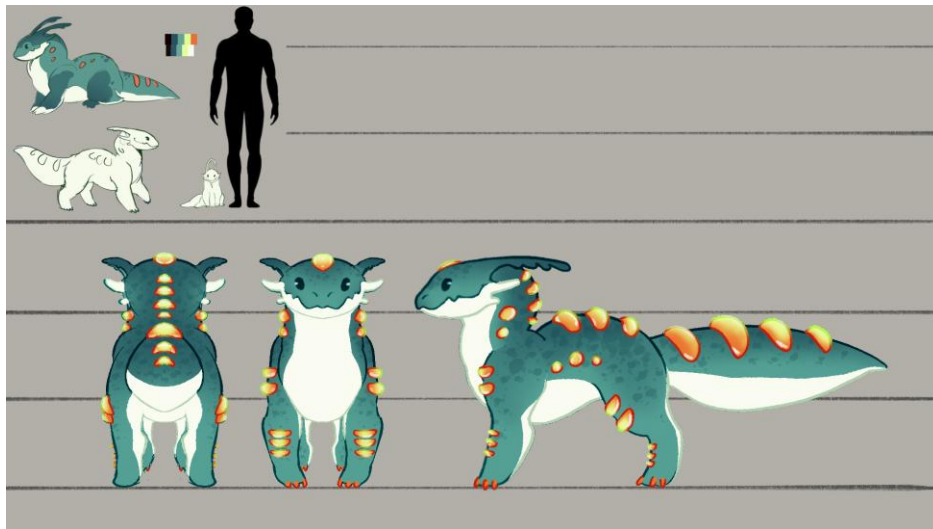


3/4 View — Face & Body



3/4 View — Body & Tail

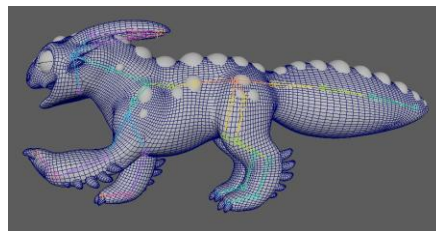
EARLY SCULPT STUDIES — FORM EXPLORATION



Early Gray Sculpt — Side Profile



Top View — Dorsal Anatomy



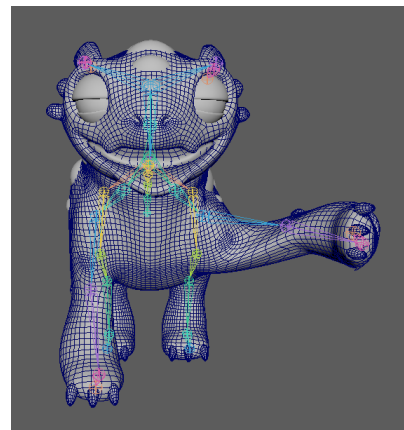
Alternate Angle



Front View

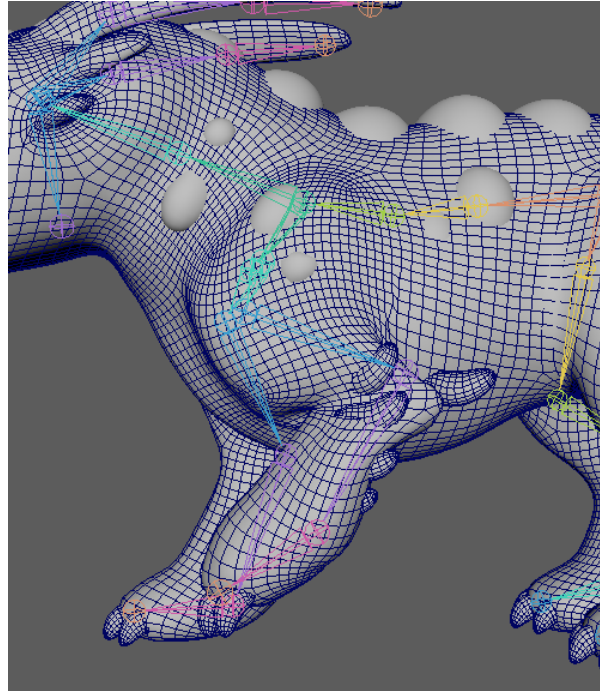


3/4 View



Head Detail

RIGGING & TOPOLOGY — MAYA



Maya — Rig Setup View

FINAL TEXTURED RENDERS — ZIGMOL



Side Render — Final PBR Textured



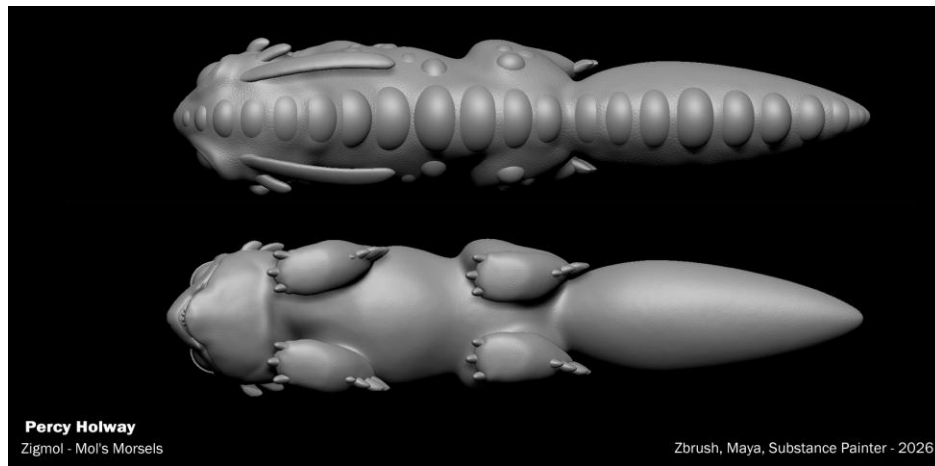
Texture Render — Orange & Blue Color Scheme



Front View — Final Model



Side View — Final Model



Top & Bottom View — Final Model

Animation Test: <https://youtu.be/tOIDHR4UnxM>

Final Animation: https://www.youtube.com/watch?v=4_Lrm9nM-JU

— — PORTFOLIO SUMMARY

Key Achievements

- ▶ Anatomical Foundation: Human hand sculpt · alligator gar anatomy study (gray sculpt to final textured render) · deer nose sculpt with custom alpha creation · herbivore creature demonstrating mastery across human and comparative animal anatomy
- ▶ Production Pipeline: High-res ZBrush sculpting → retopology → UV optimization → Substance Painter PBR baking → FiberShop/GS Curve Tools hair card system
- ▶ Technical Execution: PBR texture generation, Unreal Engine 5 shader development, environmental scene assembly, cinematic rendering achieving photorealistic presentation
- ▶ Project Management: Proactive schedule adaptation, ecosystem research foundation, parallel workflow (Independent Study + Zigmol workshop), strategic buffer utilization
- ▶ Workshop Pipeline: Zigmol — full creature pipeline from sculpt to rigged, textured, and animated asset for student short film Mol's Morsels

Portfolio: <https://percyspalooza.github.io/percyholway.portfolio/>

ArtStation: <https://www.artstation.com/percyholway>

Substance Painter: <https://youtu.be/a7bY2fN2tGg>

Hand Sculpt Video: <https://youtu.be/ctBTcpLhyzl>

Workshop Animation: <https://youtu.be/tOIDHR4UnxM>

— End of Portfolio Assessment —